

MTH441: Lab 4

Questions

1. For the `cars` dataset in R, with the response being `dist`, create a grid of λ values from $(-2, 2)$ and try to find the optimal value of λ . Does this match with the `boxcox` function in MASS?
2. Consider the `table.15` dataset in MPV. Remove `City` from the dataset and with `Mort` being the response, fit a linear regression model.
 - a. Find an appropriate transformation for the response, if necessary
 - b. Make appropriate transformations to the predictors to get the highest possible adjusted R^2 .
3. Load the dataset `test_perform.csv` that contains information on a competitive exam collected from 1000 schools.
 - a. `studs`: number of students in the school that took the exam
 - b. `y`: average score from the number of students who took the exam
 - c. `teach`: an assessment score on the quality of teachers in that school
 - d. `budget`: an assessment score on the yearly budget of the school
 - e. `extra`: as assessment score on the amount of extra-curricular activities the school encourages.

Fit a linear regression model with `y` as the response that maximizes the adjusted R^2 .